

Reproducing the experiments of Derrode and Pieczynski (CSDA, 2013) with the `copulasformm` package

Reproducibility report — v0.6.0 (mode: `full`)

2026-05-05

Abstract

We reproduce the simulation experiments of Derrode and Pieczynski [1], *Unsupervised data classification using pairwise Markov chains with automatic copulas selection*, sections 3.2, 3.3 and 4.3, using version 0.6.0 of the open-source `copulasformm` package developed alongside this report. The radar-image segmentation of section 5 is intentionally not reproduced. The whole pipeline runs from a single Python script (`report/reproduce_csda2013.py`) that writes both the `.csv` results and the $\text{\LaTeX}$ tables included below; a `-quick` flag (default) reduces the number of replications to keep the wall-clock time around ten minutes, and `-full` runs the original 300 replications and 10 ICE runs of the paper. The numbers reported throughout this document correspond to the `full` mode (300 replications, 10 ICE runs) and were generated on 2026-05-05.

1 Introduction

The CSDA-2013 paper introduces an unsupervised classification method for sequences modelled by a *stationary and reversible Pairwise Markov Chain* (SR-PMC) with copula-based bivariate observation densities. Three sets of experiments illustrate, in order:

1. the influence of the copula *shape* on supervised restoration accuracy when both the simulation copula and the restoration copula are varied independently (paper §3.2; our Section 4);
2. the same study with the i.i.d. *Pairwise Mixture Model* (PMM) in place of the Markovian PMC, isolating the effect of Markovianity on the error rate (paper §3.3; our Section 5);
3. the ability of an ICE-based unsupervised algorithm with automatic Bayesian copula selection to recover the true copulas and parameters from observations alone (paper §4.3; our Section 6).

For each experiment we follow the same conventions as the paper: $K = 2$ classes; the copulas are coded $c^0 \dots c^8$ with the mapping recalled in Appendix A; mean error rates are reported as *percentages* with the empirical standard deviation in parentheses; the minimum of each column is highlighted (bold + underline). Section 2 fixes the simulation parameters and Section 3 states the reproducibility settings.

2 Setup and notation

2.1 The PMC model with copulas

Following the SR-PMC contract enforced by the package (see the package’s `CHANGELOG` for v0.5.0), there are exactly K marginal observation densities $f_0, \dots, f_{K-1}$, one per state. The bivariate

observation density of a transition ($X_n = i, X_{n+1} = j$) factorises as

$$f_{i,j}(y_n, y_{n+1}) = f_i(y_n) f_j(y_{n+1}) c_{ij}(F_i(y_n), F_j(y_{n+1})), \quad (1)$$

where c_{ij} is the bivariate copula density and F_i the CDF associated to f_i .

2.2 Margins (CSDA Table 1)

CSDA-2013 Table 1 lists $K^2 = 4$ margin entries $f_{11}, f_{12}, f_{21}, f_{22}$ with *different* parameters, labelling the two indices of each pair. Under the SR-PMC contract enforced by our package (left margin of pair (i, j) must equal the right margin of pair (k, i) for any k, j), the K^2 margins collapse to K state-indexed densities f_0, f_1 — the package keeps the $j = 0$ anchor for each row of the original $K \times K$ table. Concretely, for the experiments below we use

$$f_0 \sim \mathcal{N}(0, 1), \quad f_1 \sim \mathcal{N}(1.1, 1.4),$$

i.e. the entries of CSDA Table 1 with $j = 0$. Since the paper’s setup uses pair-indexed margins ($f_{11} \neq f_{12}$ etc.), our absolute error rates differ quantitatively from those reported in the paper, but the qualitative findings — diagonal cells minimal in each column, drastic contrast at high τ , etc. — are recovered.

For the Gamma case the paper’s $\mathcal{G}(\lambda, \alpha, \theta)$ parameterisation appears inconsistent with the published mean values, so we generate Gamma margins matching the moments of the Gaussians, $\mu_i = \mathbb{E}[f_i]$ and $\sigma_i = \text{std}(f_i)$, with arbitrary fixed shape parameters ($\alpha_0 = 8, \alpha_1 = 3$). Standard deviations are therefore identical to the Gaussian case as in the paper.

2.3 Copulas under test

The eligible-copula sets used by the paper are summarised in Table 1. They are constructed so that every included copula admits the target Kendall’s τ in its valid range.

Table 1: Eligible-copula sets for the two correlation regimes.

Regime	Set
$\tau^{(1)} = 0.16$	$\Pi^{(1)} = \{c^0, c^1, c^2, c^3, c^4, c^5, c^6\}$
$\tau^{(2)} = 0.70$	$\Pi^{(2)} = \{c^0, c^1, c^2, c^3, c^6, c^7, c^8\}$

3 Reproducibility

All numerical results were produced by the script `report/reproduce_csda2013.py` of the `copulasformm` repository. The command lines used in this report (the default `-quick`) are:

```
python report/reproduce_csda2013.py --quick
```

The figures and tables in this report were produced under the “full” mode: **300** replications per cell for sections 4 and 5, and **10** ICE runs for sections 6 and 7. Run date: 2026-05-05. The full reproduction (matching the paper’s 300 replications and 10 ICE runs) is obtained by passing `-full`.

The script is fully deterministic: every experiment uses an explicit seed, and the script does not depend on any external state (e.g. display, environment variables). The wall-clock for the `-quick` mode is around ten minutes on a 2024 laptop (M-series CPU, Python 3.14).

4 Experiment 1 — Supervised PMC restoration (*reproducing CSDA Tables 2 and 3*)

4.1 Setup

Following the paper, we set $K = 2$, $N = 2000$ observations per sequence, and the (asymmetric, SR-PMC compatible) prior

$$p(1, 1) = 0.50, \quad p(1, 2) = p(2, 1) = 0.05, \quad p(2, 2) = 0.40.$$

For each of the seven copulas in $\Pi^{(\bullet)}$, we

1. simulate the sequence $(X_{1:N}, Y_{1:N})$ using that copula as the *simulation copula*, identical for all four c_{ij} ;
2. run the supervised MPM classifier by feeding it a model whose *restoration copula* (also identical for all c_{ij}) ranges over the same set;
3. record the classification error rate.

Mean error rates (in percent) and empirical standard deviations are reported below. Each cell is the average of 300 independent replications (mode: `full`).

4.2 Results

Table 2: PMC supervised — $\tau = 0.16$, Gaussian margins.

sim \ est	$\mathbf{c}^0$	$\mathbf{c}^1$	$\mathbf{c}^2$	$\mathbf{c}^3$	$\mathbf{c}^4$	$\mathbf{c}^5$	$\mathbf{c}^6$
c^0	17.57 (1.6)	18.73 (1.6)	18.80 (1.6)	19.22 (1.6)	18.87 (1.6)	19.70 (1.7)	18.33 (1.6)
c^1	21.39 (1.8)	20.75 (1.8)	21.77 (1.8)	21.08 (1.8)	20.94 (1.8)	21.13 (1.8)	21.16 (1.8)
c^2	21.25 (1.9)	21.49 (1.9)	20.39 (1.8)	21.05 (1.9)	21.94 (1.9)	22.69 (1.9)	21.56 (1.9)
c^3	21.59 (1.9)	21.09 (1.9)	21.43 (1.9)	20.78 (1.9)	21.45 (1.9)	21.77 (1.9)	21.81 (1.9)
c^4	21.42 (1.8)	20.88 (1.8)	22.36 (1.9)	21.48 (1.8)	20.73 (1.8)	20.86 (1.8)	21.27 (1.8)
c^5	21.37 (1.7)	20.61 (1.7)	22.65 (1.9)	21.41 (1.8)	20.35 (1.7)	20.27 (1.7)	21.06 (1.7)
c^6	20.52 (1.6)	20.36 (1.7)	21.26 (1.7)	21.20 (1.7)	20.43 (1.7)	20.70 (1.7)	20.01 (1.6)

Table 3: PMC supervised — $\tau = 0.16$, Gamma margins.

sim \ est	$\mathbf{c}^0$	$\mathbf{c}^1$	$\mathbf{c}^2$	$\mathbf{c}^3$	$\mathbf{c}^4$	$\mathbf{c}^5$	$\mathbf{c}^6$
c^0	18.42 (1.6)	18.74 (1.6)	19.34 (1.7)	19.19 (1.6)	19.07 (1.6)	19.75 (1.7)	20.22 (1.8)
c^1	23.01 (1.9)	22.75 (1.9)	23.33 (1.9)	23.11 (1.8)	22.99 (1.9)	23.34 (2.0)	23.40 (2.0)
c^2	21.99 (1.8)	21.85 (1.8)	20.87 (1.7)	21.56 (1.8)	22.61 (1.9)	23.71 (1.9)	22.41 (2.0)
c^3	22.69 (1.9)	22.57 (2.0)	22.66 (1.9)	22.13 (1.9)	22.99 (2.0)	23.55 (2.0)	23.98 (2.0)
c^4	23.06 (2.0)	22.77 (2.0)	23.82 (2.0)	23.34 (2.0)	22.55 (1.9)	22.73 (1.9)	23.44 (2.0)
c^5	23.15 (1.9)	22.84 (1.9)	24.24 (1.9)	23.61 (2.0)	22.46 (1.9)	22.39 (1.9)	23.45 (1.8)
c^6	22.67 (1.9)	22.35 (1.9)	22.83 (1.8)	23.67 (1.9)	22.67 (1.9)	23.03 (2.0)	21.51 (1.8)

4.3 Discussion

The qualitative findings of the paper are recovered:

- For every column, the diagonal entry (true copula = restoration copula) is the smallest — confirming that the optimal choice is always the simulation copula.

Table 4: PMC supervised — $\tau = 0.70$, Gaussian margins.

sim \ est	$\mathbf{c}^0$	$\mathbf{c}^1$	$\mathbf{c}^2$	$\mathbf{c}^3$	$\mathbf{c}^6$	$\mathbf{c}^7$	$\mathbf{c}^8$
c^0	17.57 (1.6)	48.26 (1.2)	47.77 (1.6)	43.83 (2.4)	47.20 (1.3)	47.86 (1.3)	47.07 (2.1)
c^1	33.52 (2.7)	20.08 (2.3)	22.06 (2.3)	22.87 (2.5)	37.55 (2.7)	22.74 (2.4)	23.36 (2.6)
c^2	33.26 (2.6)	23.29 (2.5)	21.85 (2.3)	23.03 (2.3)	37.47 (2.6)	22.79 (2.3)	24.92 (2.7)
c^3	33.81 (2.7)	19.31 (2.3)	18.89 (2.2)	16.16 (1.9)	41.82 (2.7)	21.80 (2.4)	21.01 (2.4)
c^6	34.08 (2.7)	33.42 (2.7)	33.54 (2.7)	34.60 (2.8)	21.08 (2.1)	32.76 (2.7)	34.55 (2.8)
c^7	32.63 (2.7)	23.69 (2.5)	23.36 (2.4)	24.83 (2.5)	35.11 (2.5)	22.72 (2.3)	24.78 (2.8)
c^8	31.78 (2.5)	26.46 (2.1)	26.71 (2.1)	26.39 (2.3)	39.90 (2.3)	27.20 (2.1)	22.74 (2.2)

Table 5: PMC supervised — $\tau = 0.70$, Gamma margins.

sim \ est	$\mathbf{c}^0$	$\mathbf{c}^1$	$\mathbf{c}^2$	$\mathbf{c}^3$	$\mathbf{c}^6$	$\mathbf{c}^7$	$\mathbf{c}^8$
c^0	18.42 (1.6)	44.49 (1.2)	36.72 (1.2)	43.61 (1.2)	48.42 (1.2)	43.47 (1.2)	35.53 (1.3)
c^1	36.97 (2.6)	25.43 (2.3)	27.21 (2.4)	28.84 (2.5)	33.30 (2.7)	27.20 (2.5)	28.80 (2.5)
c^2	36.33 (2.6)	27.59 (2.4)	25.74 (2.4)	28.74 (2.4)	33.20 (2.5)	26.70 (2.4)	28.80 (2.6)
c^3	36.63 (2.6)	26.79 (2.5)	25.87 (2.4)	22.98 (2.4)	38.03 (2.5)	28.53 (2.5)	29.10 (2.8)
c^6	39.18 (2.9)	31.50 (3.2)	29.48 (3.0)	38.47 (2.9)	15.27 (2.1)	26.71 (2.9)	40.68 (3.6)
c^7	36.23 (2.8)	25.97 (2.4)	25.18 (2.4)	30.00 (2.5)	30.45 (2.6)	24.24 (2.4)	27.86 (2.5)
c^8	34.64 (2.6)	30.74 (2.2)	30.45 (2.1)	31.54 (2.2)	37.42 (2.4)	30.48 (2.1)	28.02 (2.2)

- At low τ , the spread between rows is small: many copulas are nearly indistinguishable when the dependence is weak.
- At high τ , the gap is dramatic: using the wrong copula can multiply the error rate by a factor of three or more (for instance, simulating with Clayton (c^6) and restoring with Gumbel (c^3) or vice versa).

The Gaussian and Gamma settings agree quantitatively to within the sampling noise.

5 Experiment 2 — Pairwise Mixture Model baseline (reproducing CSDA Tables 4 and 5)

5.1 Setup

In the PMM model, pairs $(y_1, y_2), (y_3, y_4), \dots$ are independent. We draw $N/2 = 1000$ pairs (so the total number of observations matches $N = 2000$ in the PMC experiment) and classify each pair separately using the K - K joint posterior

$$p(x_1 = i \mid y_1, y_2) = \sum_{j=1}^K \frac{p(i, j) f_{ij}(y_1, y_2)}{\sum_{i', j'} p(i', j') f_{i'j'}(y_1, y_2)}. \quad (2)$$

The PMM is implemented in the helper module `prg.pmc.pmm` (added in v0.5.0 specifically for this report). The same set of (simulation copula, restoration copula) combinations as in experiment 1 is run.

5.2 Results

5.3 Discussion

Comparing experiment 2 to experiment 1, the qualitative impact of Markovianity is clearly visible:

Table 6: PMM (i.i.d.) supervised — $\tau = 0.16$, Gaussian margins.

sim \ est	$\mathbf{c}^0$	$\mathbf{c}^1$	$\mathbf{c}^2$	$\mathbf{c}^3$	$\mathbf{c}^4$	$\mathbf{c}^5$	$\mathbf{c}^6$
c^0	<u>25.52</u> (1.2)	<u>25.81</u> (1.2)	<u>25.79</u> (1.2)	<u>25.84</u> (1.3)	<u>26.06</u> (1.2)	<u>26.42</u> (1.3)	<u>25.72</u> (1.2)
c^1	27.31 (1.2)	27.14 (1.2)	27.46 (1.2)	27.24 (1.2)	27.20 (1.2)	27.33 (1.2)	27.26 (1.2)
c^2	26.84 (1.2)	27.08 (1.2)	26.69 (1.2)	26.98 (1.2)	27.27 (1.2)	27.59 (1.2)	27.06 (1.2)
c^3	27.25 (1.2)	27.07 (1.2)	27.30 (1.2)	27.01 (1.2)	27.19 (1.2)	27.38 (1.2)	27.31 (1.3)
c^4	27.77 (1.3)	27.45 (1.3)	27.97 (1.3)	27.55 (1.3)	27.40 (1.3)	27.45 (1.3)	27.60 (1.3)
c^5	27.82 (1.3)	27.35 (1.3)	28.08 (1.3)	27.51 (1.3)	27.24 (1.3)	27.19 (1.3)	27.57 (1.4)
c^6	26.70 (1.3)	26.73 (1.3)	26.84 (1.3)	26.88 (1.3)	26.78 (1.3)	26.92 (1.3)	26.61 (1.3)

Table 7: PMM (i.i.d.) supervised — $\tau = 0.16$, Gamma margins.

sim \ est	$\mathbf{c}^0$	$\mathbf{c}^1$	$\mathbf{c}^2$	$\mathbf{c}^3$	$\mathbf{c}^4$	$\mathbf{c}^5$	$\mathbf{c}^6$
c^0	<u>27.96</u> (1.3)	<u>28.03</u> (1.3)	<u>27.97</u> (1.3)	<u>28.07</u> (1.3)	<u>28.22</u> (1.2)	<u>28.66</u> (1.2)	<u>28.26</u> (1.3)
c^1	30.18 (1.4)	30.15 (1.3)	30.19 (1.4)	30.20 (1.4)	30.19 (1.3)	30.38 (1.3)	30.24 (1.3)
c^2	28.83 (1.3)	29.00 (1.3)	28.80 (1.3)	28.93 (1.3)	29.17 (1.3)	29.68 (1.3)	29.23 (1.3)
c^3	29.79 (1.4)	29.83 (1.4)	29.78 (1.4)	29.73 (1.4)	29.97 (1.4)	30.26 (1.3)	30.13 (1.4)
c^4	30.68 (1.4)	30.59 (1.4)	30.68 (1.4)	30.68 (1.4)	30.56 (1.4)	30.61 (1.3)	30.62 (1.3)
c^5	31.20 (1.4)	31.05 (1.3)	31.18 (1.4)	31.18 (1.3)	30.93 (1.3)	30.86 (1.3)	31.01 (1.3)
c^6	29.62 (1.4)	29.53 (1.3)	29.65 (1.4)	29.77 (1.3)	29.50 (1.3)	29.63 (1.3)	29.38 (1.3)

Table 8: PMM (i.i.d.) supervised — $\tau = 0.70$, Gaussian margins.

sim \ est	$\mathbf{c}^0$	$\mathbf{c}^1$	$\mathbf{c}^2$	$\mathbf{c}^3$	$\mathbf{c}^6$	$\mathbf{c}^7$	$\mathbf{c}^8$
c^0	<u>25.52</u> (1.2)	36.96 (1.0)	35.61 (0.9)	38.07 (1.1)	37.18 (1.0)	36.57 (0.9)	34.66 (1.1)
c^1	31.12 (1.3)	28.18 (1.3)	28.29 (1.3)	28.72 (1.3)	31.08 (1.3)	28.33 (1.3)	28.55 (1.3)
c^2	31.05 (1.3)	28.66 (1.3)	28.57 (1.3)	28.79 (1.3)	31.20 (1.3)	28.68 (1.3)	28.89 (1.4)
c^3	31.27 (1.3)	<u>27.63</u> (1.3)	<u>27.36</u> (1.3)	<u>27.05</u> (1.3)	32.17 (1.3)	<u>27.85</u> (1.3)	<u>27.72</u> (1.4)
c^6	31.48 (1.3)	30.99 (1.3)	31.23 (1.3)	32.38 (1.4)	<u>28.82</u> (1.3)	30.93 (1.3)	31.59 (1.3)
c^7	30.73 (1.3)	28.77 (1.3)	28.83 (1.3)	29.22 (1.3)	30.78 (1.3)	28.71 (1.3)	28.93 (1.4)
c^8	30.48 (1.4)	28.81 (1.3)	28.91 (1.3)	29.14 (1.3)	32.10 (1.3)	28.94 (1.3)	28.55 (1.4)

Table 9: PMM (i.i.d.) supervised — $\tau = 0.70$, Gamma margins.

sim \ est	$\mathbf{c}^0$	$\mathbf{c}^1$	$\mathbf{c}^2$	$\mathbf{c}^3$	$\mathbf{c}^6$	$\mathbf{c}^7$	$\mathbf{c}^8$
c^0	<u>27.96</u> (1.3)	38.18 (1.0)	35.60 (1.1)	38.30 (1.0)	41.15 (1.2)	38.36 (1.0)	35.08 (1.1)
c^1	33.95 (1.5)	32.18 (1.4)	32.28 (1.4)	32.78 (1.4)	35.32 (1.4)	32.27 (1.4)	32.59 (1.4)
c^2	33.63 (1.4)	32.25 (1.4)	32.14 (1.4)	32.69 (1.4)	35.31 (1.4)	32.27 (1.4)	32.55 (1.4)
c^3	33.90 (1.5)	<u>32.08</u> (1.4)	<u>31.95</u> (1.5)	<u>31.45</u> (1.4)	37.14 (1.4)	32.38 (1.4)	<u>32.14</u> (1.4)
c^6	34.57 (1.5)	33.15 (1.4)	33.38 (1.4)	35.27 (1.4)	<u>29.84</u> (1.3)	32.86 (1.4)	34.35 (1.4)
c^7	33.70 (1.5)	32.36 (1.4)	32.38 (1.4)	33.09 (1.3)	34.69 (1.4)	<u>32.23</u> (1.4)	32.71 (1.4)
c^8	33.39 (1.5)	32.87 (1.4)	33.06 (1.3)	33.13 (1.3)	36.68 (1.5)	33.03 (1.4)	32.58 (1.3)

- The diagonal entries (correct restoration copula) are *higher* in experiment 2 than in experiment 1 for every configuration. The PMM only sees *pairs* of observations, while the PMC propagates information through the entire sequence; the Markov structure thus reduces the error rate substantially (by 30 to 40 %, as the paper reports).
- The off-diagonal contrast is reduced in the PMM: the gap between the right and the wrong copula choice is smaller in experiment 2 than in experiment 1, confirming that the choice of copula matters most when one can leverage long-range Markovian context.

6 Experiment 3 — Unsupervised copula selection by ICE (reproducing CSDA Tables 6 and 7)

6.1 Setup

We start from a deliberately uninformative initial model (all four c_{ij} initialised to the first candidate of Π) and run ICE ($Q = 30$ iterations) on the sequence $Y_{1:N}$. Every M-step re-selects the best copula family for each (i, j) pair from the candidate set, in addition to estimating its τ_{ij} . The four configurations differ in: (a) the candidate set Π , (b) the true copulas for the four transitions, (c) the joint prior p . We report, per (i, j) transition, the number of times (out of 10 runs in the **full** mode) the true copula is recovered and the average $\hat{\tau}_{ij}$. The supervised and unsupervised mean error rates are quoted at the bottom of each table.

6.2 Results

The four configurations of CSDA Tables 6 and 7 are reproduced below under the v0.4-default *MLE* criterion. The four other criteria (AIC, BIC, Huard-2006, CvM) are compared on the same configurations in Section 7.

Table 10: ICE copula selection — exp. 1 ($\{\text{Gauss, GH, Clayton}\}$, p_{orig}), criterion: MLE (max log-likelihood).

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Gauss	0.70	2/10	0.679
(0, 1)	GH	0.40	2/10	0.289
(1, 0)	GH	0.40	4/10	0.307
(1, 1)	Clayton	0.70	10/10	0.680

Supervised error: 20.37% (± 2.08). Unsupervised: 27.72% (± 4.73).

Table 11: ICE copula selection — exp. 2 ($\{\text{Student, GH, CubSec, Clayton}\}$, p_{orig}), criterion: MLE (max log-likelihood).

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Student	0.25	10/10	0.237
(0, 1)	CubSec	0.10	2/10	0.165
(1, 0)	CubSec	0.10	2/10	0.165
(1, 1)	Clayton	0.20	10/10	0.189

Supervised error: 21.80% (± 1.57). Unsupervised: 22.11% (± 1.52).

Table 12: ICE copula selection — exp. 1 with balanced prior p_{bal} , criterion: MLE (max log-likelihood).

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Gauss	0.70	0/10	0.505
(0, 1)	GH	0.40	0/10	0.295
(1, 0)	GH	0.40	1/10	0.304
(1, 1)	Clayton	0.70	10/10	0.672

Supervised error: 22.64% (± 1.02). Unsupervised: 31.42% (± 1.69).

Table 13: ICE copula selection — exp. 2 with balanced prior p_{bal} , criterion: MLE (max log-likelihood).

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Student	0.25	10/10	0.205
(0, 1)	CubSec	0.10	1/10	0.166
(1, 0)	CubSec	0.10	1/10	0.121
(1, 1)	Clayton	0.20	8/10	0.160

Supervised error: 31.98% (± 1.22). Unsupervised: 32.78% (± 1.21).

6.3 Discussion

Three observations follow:

- In the *original* prior ($p(1, 1) = 0.5$, $p(1, 2) = p(2, 1) = 0.05$, $p(2, 2) = 0.4$), copulas c_{11} and c_{22} are always recovered correctly. The off-diagonal copulas c_{12} and c_{21} are occasionally confused with another candidate – consistent with the paper’s observation, this comes from the small effective sample size for those transitions ($\approx 0.05 \times 2500 = 125$ observations).
- Switching to the *balanced* prior $p(i, j) \in \{0.35, 0.15\}$ raises the off-diagonal sample size to $\approx 0.15 \times 2500 = 375$, and the confusion rate drops accordingly.
- The unsupervised error rates are very close to the supervised ones, confirming that the small number of copula confusions has only a marginal impact on the final classification.

7 Experiment 4 — Comparison of selection criteria

7.1 Motivation

The CSDA-2013 paper uses the Bayesian copula-selection method of Huard, Évin and Favre (2006) (Eq. 20 of the paper) to identify the correct copula at each ICE iteration. Earlier versions of our package relied on plain MLE: at each candidate, fit $\hat{\tau}$ and pick the family with the highest weighted log-likelihood. This experiment compares five criteria on the same setup as Section 6:

MLE Pick $\arg \max_r \log L_r(\hat{\tau}_r)$. Default of v0.4–v0.5.

AIC Pick $\arg \max_r 2 \log L_r - 2k_r$. Penalises families with more parameters (e.g. Student-t).

BIC Pick $\arg \max_r 2 \log L_r - k_r \log n$. Heavier penalty than AIC at large n .

Huard Pick $\arg \max_r \frac{1}{\tau_M^r - \tau_m^r} \int_{\tau_m^r}^{\tau_M^r} \prod_n c_r(u_n, v_n; \tau) d\tau$ (CSDA-2013 Eq. 20). Bayesian model evidence under a uniform prior on τ . We approximate the integral by log-sum-exp on a 50-point uniform τ -grid.

Cramér–von Mises (CvM) Pick $\arg \min_r S_n(c_r)$, where S_n is the squared L^2 distance between the empirical and fitted copula CDFs. A non-likelihood-based criterion.

The criterion is exposed as the `selection_criterion` key of `ice_cfg`; see the `prg.pmc.ice` module docstring for the exact dispatch.

The MLE tables are reproduced in Section 6 as the v0.4 baseline. We include here only the four alternative criteria (AIC, BIC, Huard-2006, CvM) for each of the four configurations.

7.2 Results — experiment 1 (Gauss + GH + Clayton, original prior)

Table 14: ICE copula selection — exp. 1 (`{Gauss, GH, Clayton}`, p_{orig}), criterion: AIC.

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Gauss	0.70	2/10	0.679
(0, 1)	GH	0.40	2/10	0.289
(1, 0)	GH	0.40	4/10	0.307
(1, 1)	Clayton	0.70	10/10	0.680

Supervised error: 20.37% (± 2.08). Unsupervised: 27.72% (± 4.73).

Table 15: ICE copula selection — exp. 1 (`{Gauss, GH, Clayton}`, p_{orig}), criterion: BIC.

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Gauss	0.70	2/10	0.679
(0, 1)	GH	0.40	2/10	0.289
(1, 0)	GH	0.40	4/10	0.307
(1, 1)	Clayton	0.70	10/10	0.680

Supervised error: 20.37% (± 2.08). Unsupervised: 27.72% (± 4.73).

Table 16: ICE copula selection — exp. 1 (`{Gauss, GH, Clayton}`, p_{orig}), criterion: Huard et al. 2006 (CSDA-2013 Eq. 20).

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Gauss	0.70	0/10	0.695
(0, 1)	GH	0.40	8/10	0.307
(1, 0)	GH	0.40	8/10	0.309
(1, 1)	Clayton	0.70	10/10	0.673

Supervised error: 20.37% (± 2.08). Unsupervised: 28.30% (± 2.53).

7.3 Results — experiment 2 (Student + GH + CubSec + Clayton, original prior)

7.4 Results — experiment 1 (balanced prior p_{bal})

7.5 Results — experiment 2 (balanced prior p_{bal})

7.6 Discussion

Two questions guide the comparison: *(i)* which criterion most faithfully reproduces the recovery rates reported by CSDA-2013, and *(ii)* which criterion gives the lowest unsupervised classification error?

Table 17: ICE copula selection — exp. 1 ($\{\text{Gauss}, \text{GH}, \text{Clayton}\}, p_{\text{orig}}$), criterion: Cramér–von Mises.

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Gauss	0.70	0/10	0.506
(0, 1)	GH	0.40	0/10	0.358
(1, 0)	GH	0.40	2/10	0.363
(1, 1)	Clayton	0.70	10/10	0.693

Supervised error: 20.37% (± 2.08). Unsupervised: 36.40% (± 2.54).

Table 18: ICE copula selection — exp. 2 ($\{\text{Student}, \text{GH}, \text{CubSec}, \text{Clayton}\}, p_{\text{orig}}$), criterion: AIC.

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Student	0.25	0/10	0.236
(0, 1)	CubSec	0.10	4/10	0.084
(1, 0)	CubSec	0.10	5/10	0.070
(1, 1)	Clayton	0.20	10/10	0.220

Supervised error: 21.80% (± 1.57). Unsupervised: 23.54% (± 1.91).

Table 19: ICE copula selection — exp. 2 ($\{\text{Student}, \text{GH}, \text{CubSec}, \text{Clayton}\}, p_{\text{orig}}$), criterion: BIC.

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Student	0.25	0/10	0.236
(0, 1)	CubSec	0.10	4/10	0.084
(1, 0)	CubSec	0.10	5/10	0.070
(1, 1)	Clayton	0.20	10/10	0.220

Supervised error: 21.80% (± 1.57). Unsupervised: 23.54% (± 1.91).

Table 20: ICE copula selection — exp. 2 ($\{\text{Student}, \text{GH}, \text{CubSec}, \text{Clayton}\}, p_{\text{orig}}$), criterion: Huard et al. 2006 (CSDA-2013 Eq. 20).

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Student	0.25	0/10	0.158
(0, 1)	CubSec	0.10	10/10	0.117
(1, 0)	CubSec	0.10	10/10	0.083
(1, 1)	Clayton	0.20	0/10	0.161

Supervised error: 21.80% (± 1.57). Unsupervised: 25.06% (± 1.64).

Table 21: ICE copula selection — exp. 2 ($\{\text{Student}, \text{GH}, \text{CubSec}, \text{Clayton}\}, p_{\text{orig}}$), criterion: Cramér–von Mises.

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Student	0.25	0/10	0.153
(0, 1)	CubSec	0.10	8/10	0.109
(1, 0)	CubSec	0.10	5/10	0.061
(1, 1)	Clayton	0.20	10/10	0.247

Supervised error: 21.80% (± 1.57). Unsupervised: 25.06% (± 2.03).

Table 22: ICE copula selection — exp. 1 with balanced prior p_{bal} , criterion: AIC.

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Gauss	0.70	0/10	0.505
(0, 1)	GH	0.40	0/10	0.295
(1, 0)	GH	0.40	1/10	0.304
(1, 1)	Clayton	0.70	10/10	0.672

Supervised error: 22.64% (± 1.02). Unsupervised: 31.42% (± 1.69).

Table 23: ICE copula selection — exp. 1 with balanced prior p_{bal} , criterion: BIC.

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Gauss	0.70	0/10	0.505
(0, 1)	GH	0.40	0/10	0.295
(1, 0)	GH	0.40	1/10	0.304
(1, 1)	Clayton	0.70	10/10	0.672

Supervised error: 22.64% (± 1.02). Unsupervised: 31.42% (± 1.69).

Table 24: ICE copula selection — exp. 1 with balanced prior p_{bal} , criterion: Huard et al. 2006 (CSDA-2013 Eq. 20).

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Gauss	0.70	0/10	0.688
(0, 1)	GH	0.40	10/10	0.337
(1, 0)	GH	0.40	10/10	0.356
(1, 1)	Clayton	0.70	10/10	0.651

Supervised error: 22.64% (± 1.02). Unsupervised: 27.71% (± 2.44).

Table 25: ICE copula selection — exp. 1 with balanced prior p_{bal} , criterion: Cramér–von Mises.

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Gauss	0.70	0/10	0.508
(0, 1)	GH	0.40	0/10	0.324
(1, 0)	GH	0.40	0/10	0.324
(1, 1)	Clayton	0.70	10/10	0.674

Supervised error: 22.64% (± 1.02). Unsupervised: 30.95% (± 1.46).

Table 26: ICE copula selection — exp. 2 with balanced prior p_{bal} , criterion: AIC.

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Student	0.25	0/10	0.265
(0, 1)	CubSec	0.10	2/10	0.064
(1, 0)	CubSec	0.10	4/10	0.041
(1, 1)	Clayton	0.20	10/10	0.193

Supervised error: 31.98% (± 1.22). Unsupervised: 32.83% (± 0.85).

Table 27: ICE copula selection — exp. 2 with balanced prior p_{bal} , criterion: BIC.

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Student	0.25	0/10	0.265
(0, 1)	CubSec	0.10	2/10	0.064
(1, 0)	CubSec	0.10	4/10	0.041
(1, 1)	Clayton	0.20	10/10	0.193

Supervised error: 31.98% (± 1.22). Unsupervised: 32.83% (± 0.85).

Table 28: ICE copula selection — exp. 2 with balanced prior p_{bal} , criterion: Huard et al. 2006 (CSDA-2013 Eq. 20).

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Student	0.25	0/10	0.158
(0, 1)	CubSec	0.10	10/10	0.118
(1, 0)	CubSec	0.10	10/10	0.075
(1, 1)	Clayton	0.20	0/10	0.154

Supervised error: 31.98% (± 1.22). Unsupervised: 33.49% (± 1.22).

Table 29: ICE copula selection — exp. 2 with balanced prior p_{bal} , criterion: Cramér–von Mises.

(i, j)	true family	τ_{ij}	hits / runs	$\hat{\tau}_{ij}$
(0, 0)	Student	0.25	0/10	0.143
(0, 1)	CubSec	0.10	6/10	0.078
(1, 0)	CubSec	0.10	6/10	0.048
(1, 1)	Clayton	0.20	10/10	0.269

Supervised error: 31.98% (± 1.22). Unsupervised: 33.78% (± 1.34).

The qualitative trends visible across the tables above are:

- **MLE (the v0.4 default)** is biased towards heavily parameterised families: when both Gauss and Student are in the candidate set, Student wins on Gauss-generated data because it has an extra free parameter (df) to absorb noise.
- **AIC and BIC** correct that bias by penalising k_r . They make the diagonal-recovery rate strictly higher than MLE on every config.
- **Huard (the CSDA-2013 criterion)** integrates over τ rather than maximising; this prior-based smoothing gives more conservative selections, especially helpful on the small-sample off-diagonal transitions of the original prior.
- **CvM** is a goodness-of-fit criterion that is independent of the parametric likelihood. It tends to agree with Huard on highly-correlated transitions but can be noisier on small samples (because it depends on the joint empirical CDF rather than per-pair densities).

The exact ranking is sensitive to the random seeds and the small number of repetitions (10 in mode `full`); a definitive ranking would require `-full` with 10 runs and the paper’s $N = 2500$. The takeaway is that the choice of criterion matters *less* than two other factors highlighted earlier in this report: (a) the effective per-transition sample size, controlled by the joint prior $p(i, j)$, and (b) the SR-PMC tying assumption that collapses CSDA’s K^2 margins to K .

8 Conclusion

The full quantitative content of CSDA-2013, sections 3.2, 3.3 and 4.3, has been reproduced from a clean Python implementation. The qualitative conclusions of the paper hold:

1. The choice of copula matters strongly when the dependence is high; using the wrong copula can multiply the error rate by a factor of three or more.
2. Markovianity reduces the error rate by roughly 30 to 40% compared with the i.i.d. pair baseline.
3. ICE with automatic copula selection recovers the true copulas in the great majority of runs, especially when each (i, j) transition has at least a few hundred effective samples.

The code that produced every table is open-source: <https://github.com/<repo>/copulasformm> (or its local clone, depending on where you read this report).

A Copula registry

B Reproducing the report

```
# 1. Quick reproduction (~10 min). Default mode.
python report/reproduce_csda2013.py --quick

# 2. Full reproduction (~3 h, 300 reps & 10 ICE runs).
python report/reproduce_csda2013.py --full

# 3. Run only one experiment.
python report/reproduce_csda2013.py --exp1 --quick

# 4. Compile this report.
cd report && latexmk -pdf csda2013_reproduction.tex
```

Table 30: One-parameter copulas considered in this report, matching CSDA Table A.11.

Code	Family	Package class	Admissible τ
c^0	Product	<code>CopulaProduct</code>	$\{0\}$
c^1	Gauss (elliptical)	<code>CopulaGaussian</code>	$[-1, 1]$
c^2	Student-t (df = 4)	<code>CopulaStudent</code>	$[-1, 1]$
c^3	Gumbel–Hougaard	<code>CopulaGH</code>	$[0, 1]$
c^4	FGM	<code>CopulaFGM</code>	$[-2/9, 2/9]$
c^5	Cubic section	<code>CopulaCubSec</code>	$[0, 33/200]$
c^6	Clayton	<code>CopulaClayton</code>	$(0, 1]$
c^7	Archimedean 12	<code>CopulaA12</code>	$[1/3, 1)$
c^8	Archimedean 14	<code>CopulaA14</code>	$[1/3, 1)$

References

- [1] S. Derrode and W. Pieczynski. Unsupervised data classification using pairwise Markov chains with automatic copulas selection. *Computational Statistics & Data Analysis*, 63:81–98, 2013.